

2014 A Level Paper 2 Suggested Solutions

1	Suggested Solutions (Application of Differentiation)
(i)	$x = 3t^2, y = 6t$ $\frac{dx}{dt} = 6t, \frac{dy}{dt} = 6$ $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{6}{6t} = \frac{1}{t}$ Given gradient = 0.4 $\Rightarrow \frac{1}{t} = 0.4 \Rightarrow t = 2.5$
(ii)	At $P(3p^2, 6p) \Rightarrow t = p \Rightarrow \text{gradient} = \frac{1}{p}$ Equation of tangent: $y - 6p = \frac{1}{p}(x - 3p^2)$ $y = \frac{1}{p}x - 3p + 6p$ $y = \frac{1}{p}x + 3p$ y-coordinates of $D \Rightarrow x = 0 \Rightarrow y = 3p$ $\therefore$ coordinates of $D \Rightarrow (0, 3p)$ Midpoint of $PD = \left(\frac{0 + 3p^2}{2}, \frac{3p + 6p}{2} \right)$ $= \left(\frac{3p^2}{2}, \frac{9p}{2} \right)$ Equation of locus of midpoint of PD, $x = \frac{3p^2}{2}, y = \frac{9p}{2}$ $\Rightarrow y^2 = \frac{27}{2}x$ Cartesian equation is an equation involving variables x and y only. You do not need to make y the subject since it is not required by the question.

Qn 2	Suggested Solutions (Techniques of Integration)
	$\frac{9x^2 + x - 13}{(2x-5)(x^2+9)} = \frac{3}{2x-5} + \frac{Ax+B}{x^2+9}$ $9x^2 + x - 13 = 3(x^2 + 9) + (2x-5)(Ax+B)$ when $x = 0$, $-13 = 3(9) + (-5)(B)$ $B = 8$ when $x = 1$, $9 + 1 - 13 = 3(10) + (-3)(A+8)$ $A = 3$ $\frac{9x^2 + x - 13}{(2x-5)(x^2+9)} = \frac{3}{2x-5} + \frac{3x+8}{x^2+9}$ $\int_0^2 \frac{9x^2 + x - 13}{(2x-5)(x^2+9)} dx = \int_0^2 \frac{3}{2x-5} + \frac{3x+8}{x^2+9} dx$ $= \int_0^2 \frac{3}{2x-5} + \frac{3}{2} \left(\frac{2x}{x^2+9} \right) + \frac{8}{x^2+3^2} dx$ $= \left[\frac{3}{2} \ln 2x-5 + \frac{3}{2} \ln x^2+9 + \frac{8}{3} \tan^{-1} \left(\frac{x}{3} \right) \right]_0^2$ Use law of logarithms to combine to a single logarithm term since shown form has only one logarithm term. $= \left[\frac{3}{2} \ln (2x-5)(x^2+9) + \frac{8}{3} \tan^{-1} \left(\frac{x}{3} \right) \right]_0^2$ $= \frac{3}{2} \ln \left(\frac{13}{45} \right) + \frac{8}{3} \tan^{-1} \left(\frac{2}{3} \right)$ $\therefore a = \frac{3}{2}, b = \frac{13}{45}, c = \frac{8}{3}, d = \frac{2}{3}$. Rational numbers refer to fraction.

Qn 3	Suggested Solutions (APGP)
(i)(a)	Distance run by athlete who complete first 10 stages of Version 1 $= 8 + 16 + 24 + \dots \quad (10 \text{ terms})$ $\therefore$ distance of each stage follows an AP with first term 8 and common difference 8 Total distance $= \frac{10}{2} [2(8) + (10-1)(8)] = 440 \text{ m}$
(i)(b)	Distance run by athlete who completes n stages of Version 1, $S_n = \frac{n}{2} [2(8) + (n-1)(8)]$ $= \frac{n}{2} (8 + 8n)$ $= 4n(1+n)$ $S_n \geq 5000$ ——— At least 5km, so you are solving an inequality. Make sure units are consistent (all in meters). $4n(1+n) \geq 5000$

	Using GC, When $n = 34$, $S_{34} = 4760 < 5000$ When $n = 35$, $S_{35} = 5040 > 5000$ Hence, the athlete needs to complete 35 stages to run at least 5 km.	We can use GC Table since n is an integer.
(ii)	Distance run by athlete who complete first n stages of Version 2, $D_n = 8 + 16 + 32 + 64 + \dots$ (n terms) $\therefore$ distance of each stage follows an GP with first term 8 and common ratio 2 $\therefore D_n = \frac{8}{1-2}(1-2^n) = 8(2^n - 1)$ Consider $D_n = 8(2^n - 1) = 10000$ Using GC, When $n = 10$, $D_{10} = 8184$ When $n = 11$, $D_{11} = 16376$ Distance remaining after first 10 stages of Version 2 = $10000 - 8184 = 1816$ m Distance from O to $A_{11} = \frac{16376 - 8184}{2} = 4096$ m $\begin{array}{c} 1816\text{m} \\ \overline{\hspace{10em}} \longrightarrow \\ O \hspace{10em} 4096\text{m} \hspace{10em} A_{11} \end{array}$ $\therefore$ After the athlete has run exactly 10 km of Version 2, he is 1816 m from O and is running away from O towards A_{11}.	Interpret the results: Athlete has completed 10 stages after running 8184m < 10000m. After 8184m, he is back at O to start his 11th stage. He has 1816m more to run to complete 10000m.

Qn 4	Suggested Solutions (Complex 2,3)
(a)(i)	[Out of syllabus (9758)]
(a)(ii)	[Out of syllabus (9758)]
(b)(i)	$w = \sqrt{3} - i$ $w = 2 \quad \arg(w) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$ $\therefore w = 2e^{-\frac{\pi}{6}i}$ $w^6 = \left(2e^{-\frac{\pi}{6}i}\right)^6$ $= 2^6 e^{-\pi i}$ $\equiv 64e^{\pi i}$

(b)(ii)	$\frac{w^n}{w^*} = \frac{2^n e^{-\frac{n\pi}{6}i}}{2e^{\frac{\pi}{6}i}}$ $= 2^{n-1} e^{\left(-\frac{n\pi}{6} - \frac{\pi}{6}\right)i}$ $= 2^{n-1} e^{-\frac{\pi}{6}(n+1)i}$ For $\frac{w^n}{w^*}$ to be a real number, $\arg\left(\frac{w^n}{w^*}\right) = k\pi \quad \text{where } k \in \mathbb{Z}$ $-\frac{\pi}{6}(n+1) = k\pi$ $n = -6k - 1$ $\therefore$ 3 smallest positive integer $n = 5, 11, 17$. Alternatively $\frac{w^n}{w^*} = \frac{2^n e^{-\frac{n\pi}{6}i}}{2e^{\frac{\pi}{6}i}}$ $= 2^{n-1} e^{\left(-\frac{n\pi}{6} - \frac{\pi}{6}\right)i}$ $= 2^{n-1} e^{-\frac{\pi}{6}(n+1)i}$ $= 2^{n-1} \left(\cos\left[-\frac{\pi}{6}(n+1)\right] + i \sin\left[-\frac{\pi}{6}(n+1)\right] \right)$ For $\frac{w^n}{w^*}$ to be a real number, $\sin\left[-\frac{\pi}{6}(n+1)\right] = 0$ $\Rightarrow -\frac{\pi}{6}(n+1) = k\pi \quad \text{where } k \in \mathbb{Z}$ $n = -6k - 1$ $\therefore$ 3 smallest positive integer $n = 5, 11, 17$.
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5	Suggested Solutions (Sampling theory)
(i)	[Out of syllabus (9758)]
(ii)	[Out of syllabus (9758)]

Qn 6	Suggested Solutions (PNC)
(i)	Number of teams = ${}^3C_1 {}^8C_4 {}^5C_2 {}^6C_4 = 31500$
(ii)	Case 1: Number of teams that has the midfielder brother = ${}^3C_1 {}^8C_4 {}^4C_1 {}^5C_4 = 4200$ Midfielder brother is selected. Hence 1 more midfielder needs to be selected from the remaining 4 midfielders. Attacker brother is not selected. Hence 4 attackers need to be selected from the remaining 5 attackers. Case 2: Number of teams that has the attacker brother = ${}^3C_1 {}^8C_4 {}^4C_2 {}^5C_3 = 12600$ Midfielder brother is not selected. Hence 2 midfielders need to be selected from the remaining 4 midfielders. Attacker brother is selected. Hence 3 more attacker needs to be selected from the remaining 5 attackers. Total number of teams = $4200 + 12600 = 16800$
(iii)	Method 1 Note that you are now selecting: 1 goalkeeper, 4 defenders, 2 midfielders and 4 attackers from: 3 goalkeepers, 8 defenders, 4 midfielders and 5 attackers. But one of the remaining midfielders can play as a midfielder or as a defender. Case 1: Number of teams if one of the midfielders (particular) plays as midfielder = ${}^3C_1 {}^8C_4 {}^3C_1 {}^5C_4 = 3150$ Player plays as midfielder. Hence 1 more midfielder need to be selected from the remaining 4 midfielders. Case 2: Number of teams if one of the midfielders (particular) plays as defender = ${}^3C_1 {}^8C_3 {}^3C_2 {}^5C_4 = 2520$ Player plays as defender. Hence 3 more defenders need to be selected from the 8 defenders. Player plays as defender. Hence 2 midfielders need to be selected from the remaining 3 midfielders.

Player does not play. Hence 4 defenders need to be selected from the 8 defenders.	Case 3: Number of teams if one of the midfielders (particular) does not play $= {}^3C_1 {}^8C_4 {}^3C_2 {}^5C_4 = 3150$ Player does not play. Hence 2 midfielders need to be selected from the remaining 3 midfielders. Do not forget this other case. Question did not say that the particular player needs to be selected.
	Total = $3150 + 2520 + 3150 = 8820$ Method 2 Case 1: Number of teams if one of the midfielders (particular) is considered for the role of midfielder (may or may not be selected) $= {}^3C_1 {}^8C_4 {}^4C_2 {}^5C_4 = 6300$ Case 2: Number of teams if one of the midfielders (particular) is considered for the role of defender (may or may not be selected) $= {}^3C_1 {}^9C_4 {}^3C_2 {}^5C_4 = 5670$ Note that case 1 and case 2 are not mutually exclusive i.e. there is an overlap of events, and is counted twice in the two cases. The event is when the particular player who can take two roles is not selected at all. Overcounted case: Number of teams if one of the midfielders (particular) is not selected as the midfielder and the defender $= {}^3C_1 {}^8C_4 {}^3C_2 {}^5C_4 = 3150$ No of ways = $6300 + 5670 - 3150 = 8820$

7	Suggested Solutions (Approximations)
(i)	Let X be the number of rolls, out of 10, that the fair die shows a '6'. $X \sim B\left(10, \frac{1}{6}\right)$ $P(X = 3) = 0.15505$ ≈ 0.155 (3 s.f)
(ii)	[Out of syllabus (9758)]
(iii)	[Out of syllabus (9758)]

8	Suggested Solutions (Correlation)
(a)(i)	Recall from quadratic curves that the shape for $p > 0, t > 0$ is Draw 6 points with $x > 0$ with this shape in mind.
(a)(ii)	Recall from quadratic curves that the shape for $p < 0, t > 0$ is Draw 6 points with $x > 0$ with this shape in mind.
(b)(i)	(A) $r = -0.9470$ (4 d.p.) (B) $r = -0.9749$ (4 d.p.)
(b)(ii)	Since for model (B) ($P = c \ln m + d$), the value of $r = -0.9749$ is nearer to -1, this indicates that there is a stronger negative linear correlation between variables $\ln m$ and P compared to m and P. Hence, $P = c \ln m + d$ is a better model. Equation of estimated regression line P on $\ln m$, $P = 195690 - 33660 \ln m$ $P = 196000 - 33700 \ln m \quad (3 \text{ s.f.})$
(b)(iii)	When $m = 50$, $P = 195690 - 33600(\ln 50)$ $= 64011$ $= 64000 \text{ dollars (3 s.f.)}$ Final answer to follow the accuracy given in table (3 s.f.)

9	Suggested Solutions (Hypothesis testing, t - test)
(i)	[Out of syllabus (9758)]
(ii)	[Out of syllabus (9758)]
(iii)	[Out of syllabus (9758)]

Qn 10	Suggested Solutions (Probability)
(i)(a)	Required probability = $\left(\frac{1}{10}\right)\left(\frac{2}{10}\right)\left(\frac{1}{10}\right) = \frac{2}{1000} = 0.002$
(i)(b)	Required probability = $1 - \left(\frac{9}{10}\right)\left(\frac{8}{10}\right)\left(\frac{9}{10}\right) = 1 - \frac{648}{1000} = 0.352$ <u>Alternative:</u> Case 1: $P(\star \text{ from Set 1}) = \frac{1}{10}$ Case 2: $P(\star \text{ from Set 2, no } \star \text{ from set 1}) = \left(\frac{9}{10}\right)\left(\frac{2}{10}\right)$ Case 3: $P(\star \text{ from Set 3, no } \star \text{ from set 1 and 2}) = \left(\frac{9}{10}\right)\left(\frac{8}{10}\right)\left(\frac{1}{10}\right)$ Total required probability is $\frac{1}{10} + \left(\frac{9}{10}\right)\left(\frac{2}{10}\right) + \left(\frac{9}{10}\right)\left(\frac{8}{10}\right)\left(\frac{1}{10}\right) = 0.352$
(i)(c)	Case 1: $P(\times \text{ from Set 1, } \times \text{ from Set 2, } + \text{ from Set 3}) = \left(\frac{3}{10}\right)\left(\frac{1}{10}\right)\left(\frac{2}{10}\right) = 0.006$ Case 2: $P(\times \text{ from Set 1, } + \text{ from Set 2, } \times \text{ from Set 3}) = \left(\frac{3}{10}\right)\left(\frac{3}{10}\right)\left(\frac{4}{10}\right) = 0.0036$ Case 3: $P(+ \text{ from Set 1, } \times \text{ from Set 2, } \times \text{ from Set 3}) = \left(\frac{4}{10}\right)\left(\frac{1}{10}\right)\left(\frac{4}{10}\right) = 0.016$ Required probability = $0.006 + 0.0036 + 0.016 = 0.0256$

(ii)	Required probability $= \frac{P(\text{exactly one symbol is } \star \cap \text{other two symbols are } + \text{ and } o)}{P(\text{exactly one symbol is } \star)}$ The 6 cases for the numerator (in order) are: $(+, \star, o), (+, o, \star), (o, +, \star), (o, \star, +), (\star, +, o), (\star, o, +)$ $= \frac{\left(\frac{4}{10}\right)\left(\frac{2}{10}\right)\left(\frac{3}{10}\right) + \left(\frac{4}{10}\right)\left(\frac{4}{10}\right)\left(\frac{1}{10}\right) + \left(\frac{2}{10}\right)\left(\frac{3}{10}\right)\left(\frac{1}{10}\right) + \left(\frac{2}{10}\right)\left(\frac{2}{10}\right)\left(\frac{2}{10}\right) + \left(\frac{1}{10}\right)\left(\frac{3}{10}\right)\left(\frac{3}{10}\right) + \left(\frac{1}{10}\right)\left(\frac{4}{10}\right)\left(\frac{2}{10}\right)}{\left(\frac{1}{10}\right)\left(\frac{8}{10}\right)\left(\frac{9}{10}\right) + \left(\frac{9}{10}\right)\left(\frac{2}{10}\right)\left(\frac{9}{10}\right) + \left(\frac{9}{10}\right)\left(\frac{8}{10}\right)\left(\frac{1}{10}\right)}$ The 3 cases for the denominator (in order) are: $(\star, \text{not } \star, \text{not } \star), (\text{not } \star, \star, \text{not } \star), (\text{not } \star, \text{not } \star, \star)$ $= \frac{71}{306}$
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11	Suggested Solutions (Approximations)
(i)(a)	[Out of syllabus (9758)]
(i)(b)	[Out of syllabus (9758)]
(ii)	[Out of syllabus (9758)]
(iii)	[Out of syllabus (9758)]
(iv)	[Out of syllabus (9758)]